

DRAFT

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Question #: 1

A 29-year-old man comes to the emergency department because of a 2-day history of pain and swelling in his right knee after a fall. Physical examination shows redness and swelling in the skin over the knee. The patella could not be palpated through the swelling, and a diagnosis of bursitis is made. In which of the following structures is the inflammation most likely located?

- A. Infrapatellar fat pad
- B. Prepatellar bursa
- C. Deep infrapatellar bursa
- D. Subcutaneous infrapatellar bursa
- E. Suprapatellar bursa

Question #: 2

A 20-year-old man is brought to the emergency department by paramedics 1-hour after he sustained a knife wound to the right side of his groin. The wound appears to be in the femoral triangle, and there is concern that the nerve may have been damaged. Which of the following movements will most likely be lost if the nerve in this area is injured?

- A. Adduction of thigh
- B. Extension of thigh
- C. Extension of the knee
- D. Lateral rotation of hip
- E. Flexion of knee

Question #: 3

A 17-year-old boy comes to the physician for a routine examination. Physical examination shows that stroking the medial side of the thigh results in a reflex that pulls up the cremaster. Which of

the following nerves is most likely responsible for the efferent pathway of this reflex??

- A. Ilioinguinal nerve
- B. Genital branch of genitofemoral nerve
- C. Obturator nerve
- D. Saphenous nerve
- E. Iliohypogastric nerve

Question #: 4

A 74-year-old woman is brought to the physician by her son for a routine examination. Physical examination shows strong peripheral pulses including the pulse behind the medial malleolus. Which of the following arteries is most likely palpated in this area?

- A. Anterior tibial
- B. Medial plantar
- C. Fibular
- D. Dorsalis pedis
- E. Posterior tibial

Question #: 5

A 34-year-old male was brought to the emergency department after he was involved in a motor vehicle collision that left him with a severely injured lower left limb. He is stabilized, blood samples taken for testing and cross match and immediately taken to the operating theater. Imaging studies show multiple fractures of the distal leg. A below-knee amputation was done. Which of the following cut structures belong to the lateral compartment of this region?

- A. Superficial fibular nerve, fibularis longus and tertius
- B. Deep fibular nerve, tibialis anterior and posterior
- C. Superficial fibular nerve, fibularis longus and brevis

- D.
Tibial nerve, flexor digitorum and flexor hallucis longus
- E. Common fibular nerve, fibularis longus, brevis and tertius

Question #: 6

A 59-year-old man comes to the emergency department because of a 2-hour history of a fast, irregular heartbeat. He appears diaphoretic and in respiratory distress. Physical examination shows tachycardia. In which of the following structures is the most likely location of the preganglionic cell bodies of the nerve fibers responsible for this patient's sweating?

- A. Lateral horn S2-S4
- B. Paravertebral ganglion
- C. Preaortic ganglion
- D. Lateral horn T1-L2

Question #: 7

A 25-year-old woman comes to the physician because of a 1-week history of numbness (paresthesia) of the skin in the suprapubic region of the abdomen. Two weeks ago, She underwent an emergency appendectomy. Which of the following nerves is most likely injured?

- A. Genitofemoral
- B. Iliohypogastric
- C. Subcostal
- D. Spinal nerve T10
- E. Ilioinguinal

Question #: 8

A male newborn is examined for an abnormality in his right hand as shown in the picture. Which of the following is the most likely embryological explanation for this abnormality?

- A. Disruption of the anterior-posterior pattern
- B. Failure of development of the notches between the digital rays
- C. Suppression of limb bud development in the 4th week
- D. Failure of development of one or more digital rays
- E. Loss of apoptosis of the webbing between the digits

Attachment:



for question 215082.jpg

Question #: 9

An 80-year-old woman is brought to the emergency department by ambulance after being found collapsed on her bathroom floor. Physical examination shows tenderness and limited movements in the hip joint. An x-ray is taken which is shown. Which of the following additional findings will most likely be present on physical examination of her leg?

- A. Longer and externally rotated
- B. Shorter and internally rotated
- C. Shorter and externally rotated
- D. Longer and externally rotated

Attachment:



for question 215079.jpg

Question #: 10

A 1-day-old male newborn has 'webbing' between the 2nd and 3rd toes of his left foot. Imaging studies show the phalanges of the two digits are separate and the webbing is skin only. Which of the following is the most likely diagnosis?

- A. Syndactyly
- B. Brachydactyly
- C. Polydactyly

- D. Ectrodactyly
 - E. Meromelia
-

Question #: 11

A 17-year-old male comes to the physician because of 1-week history of right knee pain. Physical examination shows right knee joint effusion. He has discomfort with varus stress test to the knee. The anterior drawer test is negative. McMurray test is negative for pain and restriction. Radiological studies are unremarkable. Which of the following structures is most likely injured ?

- A. Lateral collateral ligament injury
 - B. Medial meniscus injury
 - C. Lateral meniscus injury
 - D. Anterior cruciate ligament injury
-

Question #: 12

A 20-year-old man comes to the physician because of a laceration to the posterior lateral wrist of his right hand. Physical examination shows loss of thumb extension but normal flexion. During physical examination, what other motor deficits will most likely be observed in this patient?

- A. Loss of adduction of thumb
 - B. Weak abduction of thumb
 - C. Loss of adduction of digits 2-5
 - D. Loss of abduction of digits 2-5
 - E. Loss of opposition of thumb
-

Question #: 13

A 6-year-old boy is brought to the physician by his mother because of a structural anomaly in his hand as shown in the photograph. Which of the following is the most likely embryological explanation for this anomaly?

- A. Suppression of limb bud development in the 4th week
- B. Failure of webs to degenerate between 2 or more digits
- C. Disruption of the anteroposterior pattern
- D. Failure of development of notches between the digital rays

Attachment:



Polydactyly 2_0.jpg

Question #: 14

A 50-year-old woman is admitted to the hospital for a modified radical mastectomy because of breast cancer. During recovery, she finds difficulty reaching for items on the kitchen shelf above shoulder level. Which of the following muscles most likely lost innervation due to this surgical intervention?

- A. Levator scapulae
- B. Pectoralis major
- C. Serratus anterior
- D. Subscapularis
- E. Deltoid

Question #: 15

A 16-month-old girl is brought to the physician by her parents because of a 1-month history of bowing of the legs. Physical examination shows thickening of the wrists. Laboratory studies show elevated serum alkaline phosphatase (ALP) levels. Which of the following best describes the normal function of the vitamin most likely deficient in this patient?

- A. Increases intestinal absorption of calcium and phosphate
- B. Increases renal excretion of calcium and phosphate
- C. Involved in hydroxylation of proline in collagen
- D. Decreases synthesis of calbindins in the intestine
- E. Decreases serum calcium and phosphate levels

Question #: 16

A 60-year-old man comes to the physician because of a 2-month history of pain in his hips. He has a 4-year history of chronic renal failure and is on regular maintenance hemodialysis. Bone densitometry studies show decreased bone mineralization. Laboratory studies show low serum calcium and serum phosphate levels. Which of the following is most likely responsible for the findings in this patient?

- A. Defective calcitriol receptor
- B. Decreased activity of 25-hydroxylase
- C. Decreased formation of vitamin D in the skin
- D. Decreased activity of 1-hydroxylase
- E. Decreased secretion of parathormone

Question #: 17

A 3-year-old boy is brought to the physician because of a 3-month history of bowing of legs. Laboratory studies show low serum calcium and phosphate levels; elevated levels of 25-hydroxycholecalciferol and 1,25-dihydroxycholecalciferol (DHCC). A diagnosis of vitamin D resistant rickets is made. Which of the following is most likely abnormal in this child?

- A. 1-hydroxylase activity in kidney
- B. 25-hydroxylase activity in liver
- C. Calcitriol receptor in intestine
- D. Calcitonin secretion in response to hypercalcemia
- E. Calcium sensor in parathyroid gland

Question #: 18

A 35-year-old woman comes to the physician because of a 2-month history of bone pain and weakness. Her dietary history shows megadoses of vitamin D supplementation. Which of the following laboratory findings is most likely in this patient?

- A. Increased serum calcium levels
- B. Decreased serum 25-hydroxy vitamin D levels
- C. Decreased serum phosphate levels
- D. Decreased serum calcium levels
- E. Increased serum PTH levels

Question #: 19

Two siblings aged 3 and 5 years are brought to the physician by their parents because of a 4-week history of awkward gait. Physical examination shows the findings in the photograph. A diagnosis of a nutritional deficiency disorder is made. Which of the following is the strongest predisposing risk factor for the development of this disorder?

- A. Decreased dietary fiber content
- B. Decreased exposure to sunlight
- C. Essential fatty acid deficiency
- D. Decreased vitamin C intake
- E. Decreased dietary protein content
- F. Decreased dietary caloric content

Attachment:



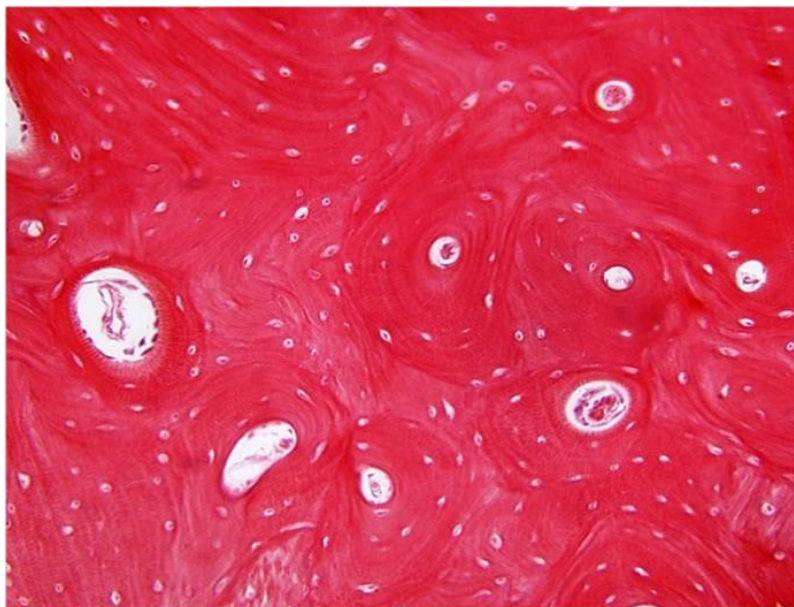
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Question #: 20

A 68-year-old man comes to the physician because of a 3-month history of pain in the right lower limb. A biopsy is collected and shown in the attached photomicrograph. Which of the following is a characteristic feature of this tissue?

- A. Matrix contains abundant type II collagen
- B. Matrix contains abundant hyaluronic acid
- C. Cells are organized as isogenous groups
- D. Matrix contains abundant elastic fibers
- E. Matrix contains abundant type I collagen

Attachment:



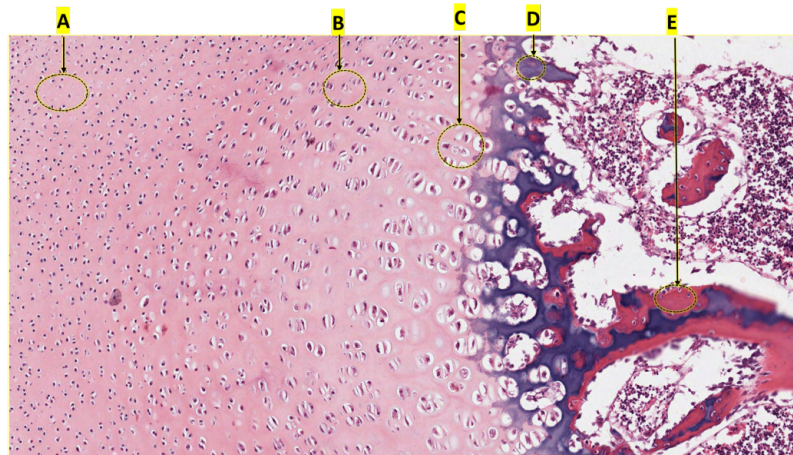
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Question #: 21

A new chemotherapy drug is shown inhibit vertical growth of the long bones. Animal models show that the drug is selectively toxic to osteoblast at the epiphyseal plate. Which of the following sites most likely contains cells that will be affected by the drug?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



3 Epiphysis.png

Question #: 22

A 43-year-old man comes to the physician because of 3 month history of difficulty in swallowing. He has smoked cigarettes for more than 20 years. Physical examination of the oral cavity shows a large mass on the epiglottis. Which of the following types of cartilage are most likely found in this region?

- A. Hyaline cartilage
- B. Elastic cartilage
- C. Fibrocartilage
- D. Fibrocartilage and hyaline cartilage
- E. No cartilage is found here

Question #: 23

A 3-year-old boy is brought to his physician by his parents because of a 2 hour history of a fall onto his arm. Physical examination shows swelling at the distal radius. Imaging studies show fracture of the distal radius with only discrete displacement of the fractured ends of the bone. Which of the following best describes the bony tissue located at the fracture site at this time?

- A. Mainly type I and II collagen
- B. Mainly proteoglycans and type II collagen
- C. Blood clot formation

D. Type II collagen and hydroxyapatite

Question #: 24

A 53-year-old woman comes to the physician because of a 3-month history of sharp pain in the right buttock and thigh. Physical examination shows no abnormalities. MRI shows a herniation of the L4-L5 intervertebral disc. Which of the following is most likely characteristic feature of the cartilage component of this structure?

- A. Matrix contains only Type II collagen fibrils
 - B. Matrix contains Type I and Type II collagen fibrils
 - C. Matrix contains abundant elastic fibers
 - D. Firmly attached perichondrium
 - E. High capability for repair
-

Question #: 25

A 5-year-old boy is brought to the physician by his mother because of a 3-month history of bowed legs. An x-ray shows kyphosis of the back, pigeon chest and bowing of the bones of both lower limbs characteristic of rickets. Which of the following best characterizes the cartilage that forms the foundation of the bones in the legs of this patient?

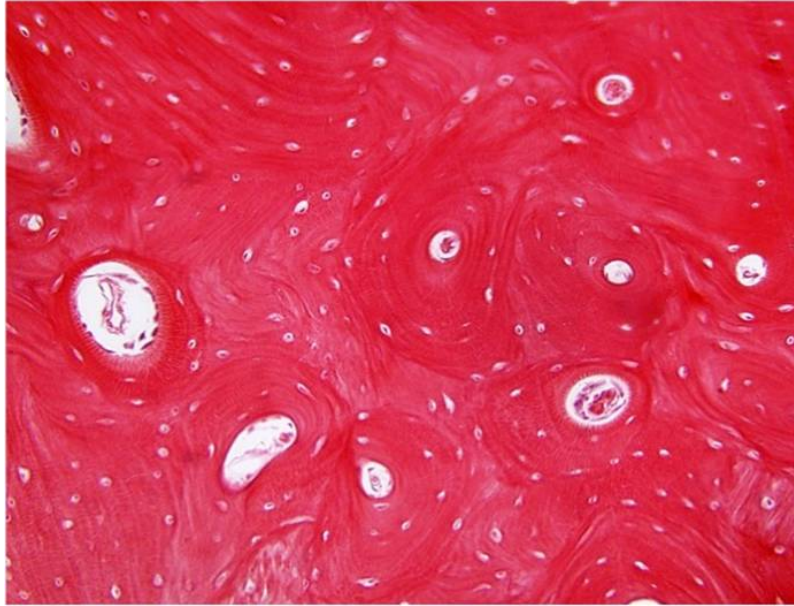
- A. Predominantly type I collagen
 - B. Isogenous groups arranged in a linear pattern
 - C. Contains a high amount of GAGs
 - D. Highly vascularized
 - E. Lacks a perichondrium
 - F. Abundance of elastic fibers
 - G. Does not undergo appositional growth
-

Question #: 26

A 2-year-old girl is brought to the physician by her parents because of a 1-week history of pain in the wrist. Physical examination shows a swollen left wrist. There are no obvious deformities. An X-ray of the wrist shows no signs of fracture and the well-defined radiolucent line on the distal radius remains intact. Which of the following is most likely a feature of this region of the bone?

- A. Contains longitudinal vascular canals
- B. Composed of five histological zones
- C. Contains transverse vascular canals
- D. Promotes appositional growth
- E. Consists of Haversian systems

Attachment:



Picture38.JPG

Question #: 27

A 75-year-old woman is brought to the emergency department by her husband because of severe pain and a left hip deformity after falling out of bed 4-hours ago. Physical examination shows tenderness and an externally rotated left lower limb. An x-ray of the pelvis and hip shows a fracture in the neck of the left femur. Bone mineral density measurement shows severe osteoporosis. Which of the following best characterizes the cartilage most likely to be affected in this disease process?

- A. Imparts tensile strength to tissues
 - B. Preponderance of elastic fibers
 - C. Rounded chondrocytes in the intermediate zone
 - D. Hypertrophied chondrocytes in the zone of hypertrophy
 - E. Found in areas of high stress
-

Question #: 28

A research group is developing a new drug to treat connective tissue disorders and wishes to investigate its effects on connective tissue elements in the body. The drug is given intravenously and x-rays are taken of different sites at various times. They discover that the drug has a predilection for bony tissue. Biopsy samples taken from some participants show destruction of specific connective tissue elements within the bone. Which of the following connective tissue fibers is most abundant in the affected tissue?

- A. Type III collagen
- B. Elastic
- C. Type IV collagen
- D. Type II collagen
- E. Type I collagen

Question #: 29

A group of researchers is studying the effects of growth hormone on height. The study involves observation of the epiphyseal growth plate in vitro. Which of the following best characterizes the zone most likely influenced by the hormone being studied?

- A. Inactive chondrocytes
- B. Calcified inter-lacunae matrix
- C. Greatly enlarged chondrocytes
- D. Chondrocytes undergoing rapid mitotic division
- E. Resorption of calcified bone-cartilage complex

Question #: 30

A 45-year-old woman comes to the physician because of a 6-month history of knee pain and stiffness. Her vital signs are within normal limits. An MRI of her knee shows thinning of the articular cartilage and early signs of osteoarthritic degeneration. Examination of a biopsy specimen of the articular cartilage shows cells that appear in isogenous groups within lacunae, surrounded by a distinct pericellular matrix, then a more fibrillar territorial matrix, and finally a more homogeneous interterritorial extracellular matrix. Which of the following cells is

most likely being observed?

- A. Chondrocytes
- B. Chondrogenic cells
- C. Chondroclasts
- D. Fibroblasts
- E. Chondroblasts